
 SHORT
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Molecular Dynamics Study of Ionic Liquid Confined in Silicon Nanopore¹

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Abstract—Molecular dynamics simulations was carried to investigate the structure and dynamics of [BMIM][PF₆] ionic liquid (IL) confined inside a slit-like silicon nanopore with pore size of 5.5 nm. It is clearly shown that the mass and number densities of the confined ILs are oscillatory, high density layers are also formed in the vicinity of the silicon surface, which indicates the existence of solid-like high density IL layers. The orientational investigation shows that the imidazolium ring of [BMIM] cation lies preferentially flat on the surface of the silicon pore walls. Furthermore, the mean squared displacement (MSD) calculation indicates that the dynamics of confined ILs are significantly slower than those observed in bulk systems. Our results suggest that the interactions between the pore walls and the ILs can strongly affect the structural and dynamical properties of the confined ILs.

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INTRODUCTION

Ionic liquids (ILs) are a novel class of organic salts with unusually low melting point, which are composed of special organic cations and different types of anions [1–3]. The interest in ionic liquids has been boosted in the past decades as a result of their tunable properties and potential applications in various technologies, such as membrane separation, air purification, lubrication, solvents, heterogeneous catalysts and reaction media [4–9]. Moreover, for some particular applications, such as photoluminescence, lithium batteries, dye-sensitized solar cells, and electrochemical sensors, etc. [10–13], one common feature is the presence of interfaces between ILs and solid surface in confinement [14]. Furthermore, most of the properties of an ionic liquid are quite different from those of the bulk, once it has been confined in the vicinity of the solid surfaces. Thus, a more detailed understanding of the behavior of confined ionic liquid is of great importance due to its relevance to these implications. Up to now, the behavior of confined ionic liquids at solid interfaces has already been studied using computational approaches. For instance, Sha et al. [15] studied 1,3-dimethylimidazolium chloride confined between graphite walls and reported the formation of a new high-melting-point solid phase, which was clearly different from that in liquid compound. The similar ordered ion distribution of ILs confined in carbon nanotube was reported by Kim et al. [16] and Dou

et al. [17]. Aparicio et al. [18] examined the structure of choline-based ILs at the interface of the graphite sheet or single-walled carbon nanotubes and found that the structure of ILs has remarkable differences in comparison with bulk ionic liquids. Shi et al. [19] and Coasne et al. [20] have found that the ILs confined into silica slit pores exhibited larger self-diffusivities than those of the bulk ILs. Meanwhile, the ILs confined inside rutile (110) slit nanopores were studied by Singh et al. [21, 22]. They found that the dynamics of ions closest to the rutile or graphite walls were significantly slower than those observed in bulk systems. For experimental studies, Gupta et al. [13] confined the [EMIM][EtSO₄] IL in silica matrix and found that the glass transition temperature (T_g) of the confined ILs was about 20°C higher than that of the bulk IL. Chathoth et al. [14] investigated the diffusivity of the [H₂NC(dma)₂][BETI] ILs. They showed that the diffusive motions became faster for a fraction of cations when the liquid was confined in a mesoporous carbon. Wu et al. [23, 24] have studied [BMIM][PF₆] confined into carbon nanotubes and silica nanopores. They found that the melting point of this ILs was clearly increased. However, in previous studies, Göbel et al. [25] reported a decrease of both the glass transition temperature and the melting point of 1-ethyl-3-methylimidazolium-based ionic liquids confined into mesoporous silica by the sol-gel technique. These studies have provided important insights into the properties of ILs at the interfaces with solid walls,

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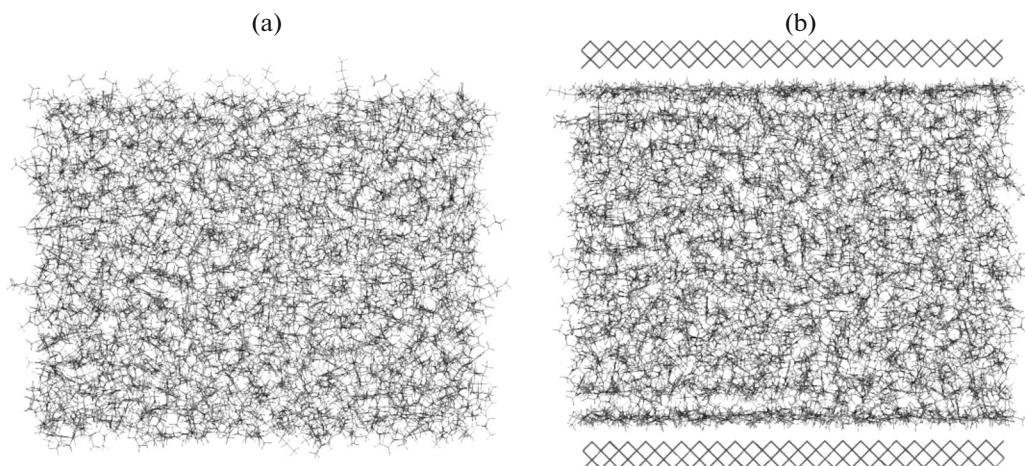


Fig. 1. Representative simulation snapshots of the [BMIM][PF₆] IL. (a) Bulk IL, (b) IL confined inside a silicon slit pore.

when the ILs are confined inside pores of simple geometries. The unusual behavior of the confined ILs, as well as their physical and chemical interactions with solid surfaces, is hardly understood at present and requires further investigation.

In this work, we have performed molecular dynamics (MD) simulations of the IL [BMIM][PF₆] confined between two parallel silicon walls with a pore size of 5.5 nm. The structural and dynamic properties were evaluated for the confined ILs. The MD calculation indicates that the solid-like layers of [BMIM][PF₆] IL were formed in the vicinity of silicon surface. Also, the orientation studies show that the imidazolium ring of ILs lie in the plane of the silicon surface.

SIMULATION METHODOLOGY AND MODEL

The [BMIM][PF₆] ionic liquid was treated as a systematic OPLS-AA force field as developed by Lopes et al. [26], which was successfully used to simulate the confined ionic liquids [16, 17, 27, 28]. The simulation box contained two silicon sheets with dimensions $X = Y = 6.52$ nm, the flat silicon surface includes double layers of silicon sheet with a total of 2304 atoms. The size of the silicon sheets in the x and y directions is 6.51 nm. In our simulation, the IL was placed inside the silicon slit pore (5.5 nm), which was obtained at pore loading [BMIM][PF₆] IL $\rho = \rho_{\text{bulk}}$ and contain 654 [BMIM][PF₆] ion pairs. The pore size was defined as the minimum surface-to-surface distance between the two silicon walls at z direction. In the simulations, we conducted long runs of 20 ns to make sure that the results are statistically converged. The non-bonding interactions were calculated using a 1.5 nm atom-based cut-off, correcting for the long-range electrostatics using the Ewald summation method (Particle Mesh Ewald “PME” approximation). The MD simulations were performed at 300 K with

GROMACS MD package [29] version 5.0.6 in the canonical ensemble (constant NVT).

RESULTS AND DISCUSSION

IL density. As shown in Fig. 1, comparing with bulk systems (Fig. 1a), the layer structures (Fig. 1b) are formed near the silicon surface. We present mass density profiles to the silicon surface along the z axis in Fig. 2. The simulation results indicate that the mass densities of the ILs are oscillatory, several separated high density layers are formed near the silicon surface. This is in agreement with the previous studies [14, 16, 17, 24–28, 30]. Also, as shown in Fig. 2, the heights of the first and second peaks close to the silicon interface are about 67 and 21% higher, respectively, than that of the bulk IL, which indicates the existence of solid-like layers in the vicinity of silicon surface. The oscillations in density reduce in amplitude towards the center of the pore, while the IL density also converges swiftly to the density of bulk IL. The formation of such a dense phase layer is mainly due to a strong interfacial effect between the silicon surface and the confined [BMIM][PF₆] ion pairs [31].

Number density. We present the number density profiles of the confined IL along the z direction in Fig. 3. Similar to the mass density distribution profiles, the number density profiles of [BMIM]⁺ cations (Fig. 3a) and [PF₆][−] anions (Fig. 3b) exhibit pronounced oscillations. At the IL-silicon interface, the number density distributions of ions exhibit two peaks, which are characterized by approximately equal amplitude at 0.4 and 0.3 nm, respectively. This directly corresponds to the change of mass density. Furthermore, cations as well as anions have the highest density near the pore surfaces, with noticeable layering behavior observed throughout the system. Due to strong interfacial interactions, the confined ions were aligned along the silicon pore walls to form polar

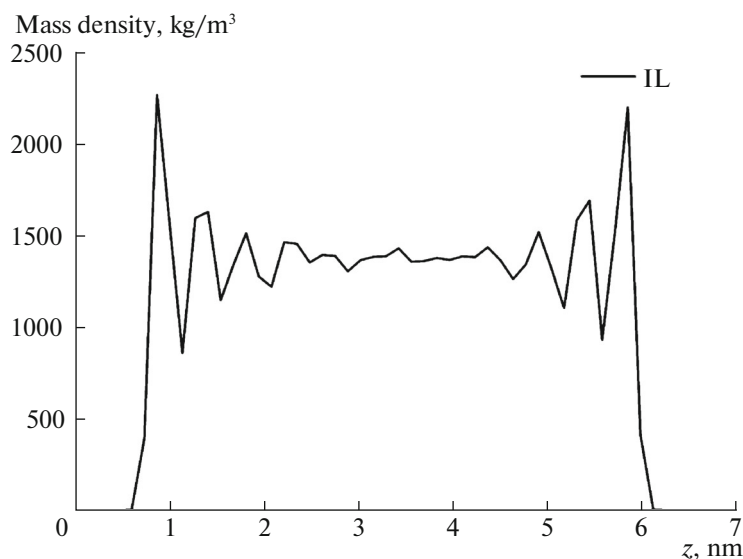


Fig. 2. Mass density profiles along the z axis for the [BMIM][PF₆] confined inside a silicon nanopore.

domains [30], which are responsible for layer structures and mass or number density increase of the confined IL at IL–solid interface [30, 31].

Orientational ordering distributions. The orientational ordering distributions of [BMIM]⁺ cations in simulated systems can be described by second Legendre polynomials as $\langle P_2(\cos \theta) = 1/2(3\cos^2\theta - 1) \rangle$, where θ denotes the angle between surface normal (z axis) and the vector of the imidazolium ring normal. In Fig. 4, we analyzed the distributions of imidazolium ring at IL–silicon interfaces and obtained a clearer cation configuration of orientational preferences. The position $z = 0.9$ and 5.7 nm corresponds to the edge of the first IL layer. When z is less than 0.9 or greater than 5.7 nm, the $P_2(\cos \theta)$ value of the imidazolium ring normal is about 0.93 , indicating that the imidazolium ring lies flat on the silicon surface (Fig. 1b). When z is beyond 0.9 and less than 5.7 nm, toward to the center of the box, the orientation of the imidazolium ring becomes inconspicuous and finally stochastic. This is mainly due to the strong interactions between the cations and the silicon surface. Imidazolium IL may be applied to form more stable and well-regulated layers at silicon surfaces [15, 17, 30, 32]. As a result, the imidazolium ring plane of the confined ILs preferably takes parallel orientation at the silicon surface, which is much more energetically favorable than any other orientations. These results are important for the understanding of the ionic packing in nanoconfinement, and may explain the observed experimental differences in imidazolium ring distributions at IL–solid interface.

Dynamical properties. In Fig. 5, we show the dynamics of the confined [BMIM][PF₆] IL by calculating the mean square displacements (MSD) of

z direction as a function of time for cations (Fig. 5a) and anions (Fig. 5b). As shown in Fig. 6, the MSD increases rapidly at very short times and then exhibits a plateau at larger times, and the mobility of the confined ions increases monotonically. Our results also indicate that the dynamics of the confined cations and

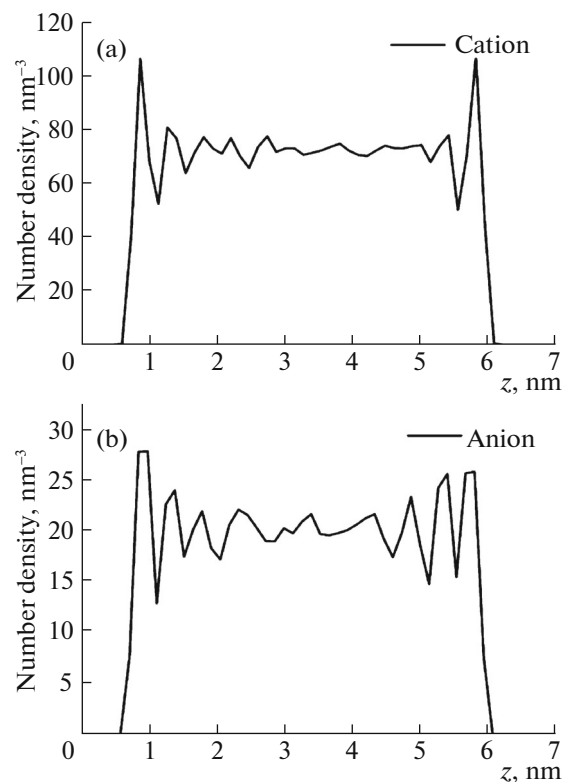


Fig. 3. Number density profiles along the z axis for the IL confined inside a silicon nanopore; (a) cation, (b) anion.

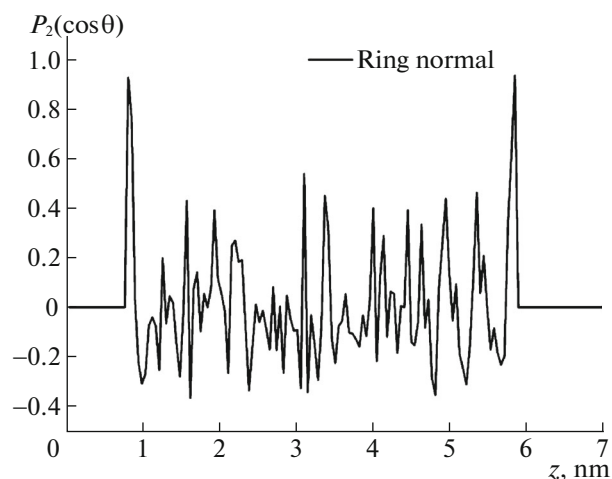


Fig. 4. Average $P_2(\cos\theta)$ of the angles between the normal of imidazole ring plane and the silicon pore walls (surface normal of z axis).

anions are significantly slower than those observed in bulk systems. Thus, as observed in previous studies [22, 33–35], [BMIM][PF₆] confined in the silicon nanopore is found to diffuse slower than the corresponding bulk IL. This is mainly because of the confinement of this very narrow pores [36], in the region near pore walls, IL molecules are condensed by the interaction with the silicon pore walls, the silicon wall retards IL movement; on the other hand, the layering structure is formed, the mobility of ILs is reduced [21, 33], this reduction in mobility might be responsible for the decrease in mobility of the confined ILs [19, 34, 37]. These results indicate the strong interactions between the ions and the surface.

CONCLUSION

The structural and dynamical properties of the ionic liquid [BMIM][PF₆] inside a silicon slit nanopore were studied using molecular dynamics simulations. Compared to the bulk ILs, the [BMIM][PF₆] ionic liquids formed layered structure along the pore walls. The simulations also indicate that the cations close to the pore surface tend to align with their imidazolium ring approximately parallel to the surface. Moreover, the ILs have the higher mass and number density peaks near the silicon surface. The dynamics of ILs inside a slit-like silicon pore is significantly slower than for the bulk IL. This can be due to strong interactions between the ions and the silicon solid surface. These findings are important for understanding the modification or the microstructure of the confined ionic liquids. We hope that this study may inspire further computational and experimental research on ILs near differently featured solid walls and for the development of practical applications of confined ILs.

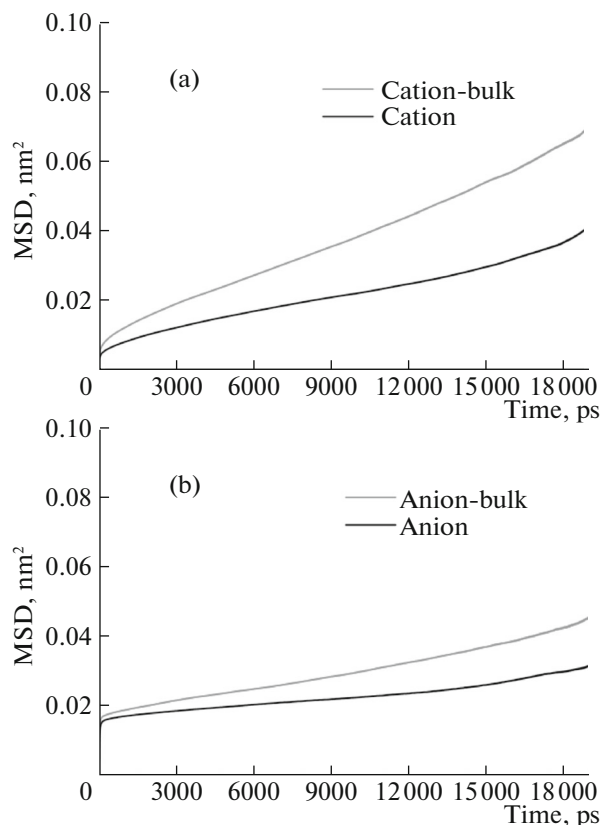


Fig. 5. Mean square displacement of cations (a) and anions (b) inside the silicon nanopore.

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